Southern New Hampshire University

CS 255 Business Requirements Document Template

Complete this template by replacing the bracketed text with the relevant information.

This template lays out all the different sections that you need to complete for Project One. Each section has guiding questions to prompt your thinking. These questions are meant to guide your initial responses to each area. You are encouraged to go beyond these questions using what you have learned in your readings. You will need to continually reference the interview transcript as you work to make sure that you are addressing your client's needs. There is no required length for the final document. Instead, the goal is to complete each section based on your client's needs.

Tip: You should respond in a bulleted list for each section. This will make your thoughts easier to reference when you move into the design phase for Project Two. One starter bullet has been provided for you in each section, but you will need to add more.

System Components and Design

Purpose

What is the purpose of this project? Who is the client and what do they want their system to be able to do?

- Create a system for DriverPass that provides driver education
    - Online exams
    - Training with instructors
    - Scheduling
    - Payment methods
- DriverPass wants a cloud-based, secure, and easily accessible system

System Background

What does DriverPass want the system to do? What is the problem they want to fix? What are the different components needed for this system?

- DriverPass would aim to fix the 65% of students that fail the DMV driving tests due to no preparation this will be done by
    - Having structured training with packages
        - 6 hours, 8 hours, 12 hours
    - Scheduling of lessons
    - Track progress
    - Have several roles for users that being students instructor and admins

Objectives and Goals

What should this system be able to do when it is completed? What measurable tasks need to be included in the system design to achieve this?

- Students should be able to view their lesson progress, test scores
- Make payments online
- Schedule driving lessons and practice tests

- Administrators should also be able to manage customer accounts, process payments and have financial reports.

Requirements

Nonfunctional Requirements

In this section, you will detail the different nonfunctional requirements for the DriverPass system. You will need to think about the different things that the system needs to function properly.

Performance Requirements

What environments (web-based, application, etc.) does this system need to run in? How fast should the system run? How often should the system be updated?

- Web Based system that can be accessed from phones and browsers
- Should load data under 5 seconds
- Handle anywhere from 350-750 users
- Secuirty monitoring with consistent updates

Platform Constraints

What platforms (Windows, Unix, etc.) should the system run on? Does the back end require any tools, such as a database, to support this application?

- Some possible constraints are
    - How the web application will be supported and look on mobile devices
    - What database or server type is to be used to store user data and scheduling services

Accuracy and Precision

How will you distinguish between different users? Is the input case-sensitive? When should the system inform the admin of a problem?

- Differentiate users through email and passwords
- Case sensitive will matter even on passwords
- Admins should be informed of payment issues, scheduling conflicts and errors

Adaptability

Can you make changes to the user (add/remove/modify) without changing code? How will the system adapt to platform updates? What type of access does the IT admin need?

- The user should be able to be removed added or modified through an administrators side and should not change the code
- The system should use API integrations and cloud deployment to allow for more headroom
- IT admins should have full access to the systems configurations

Security

What is required for the user to log in? How can you secure the connection or the data exchange between the client and the server? What should happen to the account if there is a "brute force" hacking attempt? What happens if the user forgets their password?

- The user will be requires to long using a secure username and password
- Encrypting data during exchanges would be the best way to go between client and server
- If there is a brute force hacking attempt for example after 3-5 incorrect passwords are entered the user should require additional verification and be locked until the account is verified
- Users should be able to reset passwords through email

Functional Requirements

Using the information from the scenario, think about the different functions the system needs to provide. Each of your bullets should start with "The system shall . . ." For example, one functional requirement might be, "The system shall validate user credentials when logging in."

- The system shall allow users to create and manage their account
- The system will allow for users to perform practice tests and view lessons
- The system will allow for users to have a secure payment transaction
- The system shall notify users of upcoming lessons and lessons via email or sms
- The system will create administrators logs with users schedule

User Interface

What are the needs of the interface? Who are the different users for this interface? What will each user need to be able to do through the interface? How will the user interact with the interface (mobile, browser, etc.)?

- The different users are students, instructors and administrators
- Student interface:
    - Register/login
    - View lessons and schedule lessons
    - View and manage driving lessons
    - Take practice tests
    - Some form of payments
    - Access learning materials
- Instructor Interface:
    - View scheduled lessons
    - Update availability
    - Provide feedback to students
    - Reach out to administrators
- Administrators interface:
    - Manage accounts
    - Oversee scheduling and notify of possible conflicts
    - Ensure payments have been process and possible refunds
    - Have reports be generated for both financial and students

Assumptions

What things were not specifically addressed in your design above? What assumptions are you making in your design about the users or the technology they have?

- I am assuming the user will have internet access and a web browser that is fast
- Customers/students will have proper scheduling information
- Payment will be handled third part – PayPal
- Users will understand the UI

Limitations

Any system you build will naturally have limitations. What limitations do you see in your system design? What limitations do you have as far as resources, time, budget, or technology?

- No in person help everything is essentially online – this means payment
- Internet connectivity and resource management therefore users must have basic internet access and understanding
- Language support being mainly English
- Data storage will either be limited and possibly terminated once the student passes their DMV test.

Gantt Chart

Please include a screenshot of the GANTT chart that you created with Lucidchart. Be sure to check that it meets the plan described by the characters in the interview.

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Gantt Chart For DriverPass System Development

